SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA17	**Course Title:**	Artificial Intelligence
CO Assessed:	CO1 – Apply (BL3)	**Assessment:**	Assessment Tool 1 – Analytical Problem Solving
Weightage:	35% of CIA	**Duration / Marks:**	60 mins
CO1 Statement:	Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)		
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Section / Batch:	Slot A	**Date:**	5-10-2026

Code of Conduct

I K. Madhulika , 192524190 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _Madhulika_

Instructions

- Read each question carefully before answering.
- Answer all four questions. Show every step: state your assumptions, the method you choose and why, your calculations, and finally interpret the result for the agent or AI system in the problem.
- Questions are drawn from Foundations and History of AI, Intelligent Agents (Agents and Environments) and the Concept of Rationality.
- Prepare digital Write up units and clear final answers (marks, minutes, percentage points, expected utility); present tables and working neatly.
- Marks will be awarded using the Analytical Problem Solving rubric: problem understanding, method selection, solution process, accuracy of calculation and interpretation of results.
- Upload in Github and GCR (Question paper with regNo, Name, Answers and rubrics and evalution page at last).
- Duration: 1 hour (15 minutes per question).

Section / Topic	Focus	Question Nos.
1. Foundations and History of AI	Definitions of AI (four approaches), Turing Test, eras and milestones in the history of AI	**Q3**
2. Intelligent Agents: Agents and Environments	Agent function and program, PEAS description, task environment properties, agent architectures	**Q1, Q2**
3. Concept of Rationality	Performance measure, percept sequence, expected utility, rational versus omniscient agent	**Q1, Q4**

Question 1: Two-Room Vacuum World

Topic: Agents and environments, concept of rationality | 15 minutes

A vacuum-cleaner agent lives in a world with two squares, A and B. At each time step it perceives its own location and whether that square is Dirty or Clean. Its actions are Suck, Right, Left and NoOp. The agent starts in square A, and both squares are initially Dirty. Once a square is cleaned, dirt never reappears.

Performance measure over a lifetime of T = 6 time steps: +1 for every clean square at the end of each time step, and a penalty of p for every Left or Right move (take p = 1 unless told otherwise).

- **Agent X (simple reflex):** if the current square is Dirty, Suck; otherwise move to the other square.
- **Agent Y (reflex with internal state):** if the current square is Dirty, Suck; otherwise, if it remembers that both squares have been cleaned, NoOp; otherwise move to the other square.

(a) Write the PEAS description (Performance measure, Environment, Actuators, Sensors) of this agent from the data given. **[4 Marks]**

(b) Trace Agent X for all 6 time steps in a table (action, clean squares at end of step, move penalty, step score) and find its total score for p = 1. **[5 Marks]**

(c) Trace Agent Y in the same way and find its total score for p = 1. **[4 Marks]**

(d) Derive the total score of X and of Y as functions of p. For which values of p is Y strictly better than X? **[3 Marks]**

(e) Which agent is rational? Interpret your result: explain how the performance measure and the percept history decide rationality, and why X cannot match Y even though both see the same percepts at each step. **[4 Marks]**

Question 2: Task Environment Analysis and Agent Selection

Topic: Agents and environments (PEAS, environment properties, agent types) | 15 minutes

Three AI agents are to be deployed in the following task environments:

- **E1:** An autonomous taxi driving through city traffic.
- **E2:** A part-picking robot that uses a camera to pick defective parts off a moving conveyor belt and place them in bins.
- **E3:** A chess-playing agent in a tournament game played with a clock.

Difficulty score convention: for each of the six properties below, the harder value scores 1 point and the easier value scores 0. A semi-dynamic environment scores 0.5 on the Static/Dynamic property.

Property	Easier value (0)	Harder value (1)
Observability	Fully observable	Partially observable
Number of agents	Single agent	Multi-agent
Determinism	Deterministic	Stochastic
Episodic / Sequential	Episodic	Sequential
Static / Dynamic	Static	Dynamic (semi-dynamic = 0.5)
Discrete / Continuous	Discrete	Continuous

(a) Classify each environment on all six properties with a one-phrase justification for each. **[9 Marks]**

(b) Compute the difficulty score of E1, E2 and E3 and rank them from hardest to easiest. **[4 Marks]**

(c) For the hardest and the easiest environment, choose the most suitable agent architecture (simple reflex, model-based reflex, goal-based, utility-based, learning) and justify the choice using your classification. **[4 Marks]**

(d) Interpret: explain how the properties of an environment, not just its domain, decide how complex the agent program must be. **[3 Marks]**

Question 3: Turing Test Data Across the History of AI

Topic: Foundations and history of AI (Turing Test, acting humanly, eras of AI) | 15 minutes

In a Turing-test style study, judges held five-minute text conversations with an unknown party and then said whether it was a human or a machine. Each system below faced 50 independent interrogations in which it was the machine. The table shows how many times the judge correctly identified it as a machine.

System	Era / technique	Interrogations	Correctly identified
ELIZA	1960s, keyword pattern matching	50	47
Chatbot B	2010s, scripted rules plus statistical NLP	50	36
System C	2020s, large neural language model	50	29

Reference: Turing (1950) predicted that an average interrogator would have no more than a 70% chance of making the right identification after five minutes. A judge who is purely guessing is right 50% of the time.

(a) Compute the correct-identification rate and the fooling rate (percentage of interrogations in which the machine was taken for a human) for each system. **[6 Marks]**

(b) Decide which systems satisfy Turing's 70% criterion. State the method you use and how far each system is from the 50% chance level. **[5 Marks]**

(c) Compute the improvement in fooling rate from ELIZA to System C in percentage points and as a ratio. Also compute the improvement from Chatbot B to System C in percentage points. **[5 Marks]**

(d) Interpret: the Turing Test measures "acting humanly". Explain why a high fooling rate does not prove that System C is thinking or acting rationally, using one idea from the foundations of AI (for example the four approaches to defining AI, or the Chinese Room argument). **[4 Marks]**

Question 4: Rational Route Choice under Uncertainty

Topic: Concept of rationality (expected performance, rational versus omniscient agent) | 15 minutes

A rational taxi agent must reach the airport. The deadline is 55 minutes. Its performance measure (utility) for a trip is:

$$\mathbf{U = 100 - T - 20 \cdot L}$$

where T is the travel time in minutes and L = 1 if the taxi arrives after the 55-minute deadline, otherwise L = 0.

Route	Normal time (min)	Probability of disruption	Time if disrupted (min)
R1: Highway	30	0.30 (accident)	70
R2: City road	45	0.10 (signal failure)	60
R3: Ring road	50	0 (never disrupted)	50

(a) Compute the utility of each route in each outcome (normal and disrupted), clearly marking the late trips. **[4 Marks]**

(b) Compute the expected utility of R1, R2 and R3 and state which route the rational agent should choose. **[5 Marks]**

(c) A new percept arrives: a radio alert reports a major accident on the highway, raising the probability of disruption on R1 to 0.80. Recompute the expected utility of R1 and re-decide the route. **[4 Marks]**

(d) Suppose the agent chose the route in part (b) and the disruption did occur. Compute the utility actually obtained and state whether the agent was irrational. Distinguish between a rational agent and an omniscient agent. **[4 Marks]**

(e) Interpret: explain why rationality depends on the percept sequence and the performance measure, and why part (c) changes the decision although the roads themselves did not change. **[3 Marks]**

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary (100% of marks)	Level 3 – Proficient (75%)	Level 2 – Developing (50%)	Level 1 – Beginning (25%)
Problem Understanding	**20%**	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	**20%**	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	**25%**	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	**20%**	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	**15%**	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Q. No. / Criteria	Problem Understanding (4)	Method Selection (4)	Solution Process (5)	Accuracy of Calculation (4)	Interpretation of Results (3)	Sub. Total (20)	Total %
1							
2							
3							
4							
Total							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

CO1 AT1 – Analytical Problem Solving

Name: K. Madhulika

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CSA 1716 – Artificial Intelligence

1. Two-Room Vacuum World

The vacuum agent starts in A, and both A and B are dirty. The performance measure gives +1 for each clean square at the end of every step and a penalty of p for every Left/Right movement. For the given case, p = 1.

(a) PEAS Description

Component	**Description**
Performance Measure	+1 for every clean square at the end of each time step; −p for every Left/Right movement
Environment	Two squares A and B; initially both are dirty
Actuators	Suck, Left, Right, NoOp
Sensors	Perceives its current location and whether the current square is Dirty or Clean

The agent uses its percepts to decide whether to clean the current square or move to the other square.

(b) Trace Agent X

Agent X is a simple reflex agent:

- If current square is Dirty → Suck
- Otherwise → move to the other square.

For p = 1:

Step	Location	Percept	Action	Clean squares at end	Move penalty	Step score
1	A	Dirty	Suck	A	0	+1
2	A	Clean	Right	A	−1	0
3	B	Dirty	Suck	A, B	0	+2
4	B	Clean	Left	A, B	−1	+1
5	A	Clean	Right	A, B	−1	+1
6	B	Clean	Left	A, B	−1	+1

Total score

1+0+2+1+1+1=6

Therefore, Agent X's total score = 6.

(c) Trace Agent Y

Agent Y has an internal state. Once it remembers that both squares have been cleaned, it performs No instead of continuing to move.

Step	Location	Percept	Action	Clean squares at end	Move penalty	Step score
1	A	Dirty	Suck	A	0	+1
2	A	Clean	Right	A	−1	0
3	B	Dirty	Suck	A, B	0	+2
4	B	Clean	NoOp	A, B	0	+2
5	B	Clean	NoOp	A, B	0	+2
6	B	Clean	NoOp	A, B	0	+2

Total score

1+0+2+2+2+2=9

Therefore, Agent Y's total score = 9.

(d) Score as a function of p

Agent X

The movement penalties occur at steps 2, 4, 5 and 6.

Without movement penalties:

$1+1+2+2+2+2=10$

There are four movements, so:

$X(p)=10-4p$

Agent Y

Agent Y makes only one movement at step 2.

$Y(p)=10-p$

For Y to be strictly better:

$10-p>10-4p$

$3p>0$

$p>0$

Therefore, Agent Y is strictly better than Agent X whenever the movement penalty is greater than 0.

(e) Which agent is rational?

Agent Y is the rational agent because it considers its percept history using internal state.

Agent X reacts only to the current percept. Even after both squares are clean, it sees a clean square and therefore keeps moving between A and B.

Agent Y remembers that both squares have already been cleaned. Therefore, it chooses NoOp, avoiding unnecessary movement penalties.

The rationality of an agent depends on the performance measure and the percept sequence, not simply on the current percept. Since movement has a penalty, continuously moving reduces performance.

Although both agents receive the same percepts at each step, Agent Y uses its internal state/percept history, while Agent X cannot remember previous percepts.

Final conclusion:

Agent Y is more rational because it achieves higher performance.

2. Task Environment Analysis and Agent Selection

The question gives three environments: autonomous taxi, conveyor-belt robot and chess. The six properties are observability, number of agents, determinism, episodic/sequential, static/dynamic and discrete/continuous.

(a) Classification of environments

E1 – Autonomous Taxi

Property	**Classification**	**Justification**
Observability	Partially observable	Taxi cannot know everything about traffic/pedestrians
Agents	Multi-agent	Other vehicles and pedestrians act independently
Determinism	Stochastic	Traffic conditions and other actions are uncertain
Episodic/Sequential	Sequential	Current actions affect future situations
Static/Dynamic	Dynamic	Traffic changes continuously
Discrete/Continuous	Continuous	Position, speed and movement vary continuously

E2 – Part-Picking Robot

Property	Classification	Justification
Observability	Partially observable	Camera may not provide complete information
Agents	Single-agent	The robot performs the task independently
Determinism	Stochastic	Defect detection and object movement may be uncertain
Episodic/Sequential	Sequential	Each action affects the next picking situation
Static/Dynamic	Dynamic	Conveyor belt is continuously moving
Discrete/Continuous	Continuous	Objects and robot movement occur continuously

E3 – Chess-playing Agent

Property	Classification	Justification
Observability	Fully observable	Complete board position is visible
Agents	Multi-agent	Two players compete
Determinism	Deterministic	A legal move has a predictable board result
Episodic/Sequential	Sequential	Each move affects future moves
Static/Dynamic	Semi-dynamic	Board is static but the clock changes
Discrete/Continuous	Discrete	Moves and board states are discrete

(b) Difficulty Score

According to the question, the harder value receives 1 point, while the easier value receives 0; semi-dynamic receives 0.5.

E1 – Autonomous Taxi

- Partially observable = 1
- Multi-agent = 1
- Stochastic = 1
- Sequential = 1
- Dynamic = 1
- Continuous = 1

E1=6

E2 – Part-Picking Robot

- Partially observable = 1
- Single agent = 0
- Stochastic = 1
- Sequential = 1
- Dynamic = 1
- Continuous = 1

E2=5

E3 – Chess

- Fully observable = 0
- Multi-agent = 1
- Deterministic = 0
- Sequential = 1
- Semi-dynamic = 0.5
- Discrete = 0

E3=2.5

Ranking:

Rank	Environment	Difficulty
1	**E1 – Autonomous Taxi**	**6**
2	**E2 – Part-Picking Robot**	**5**
3	**E3 – Chess**	**2.5**

Therefore:

E1>E2>E3

(c) Agent architecture

Hardest: E1 – Autonomous Taxi

The most suitable architecture is a Utility-Based Agent.

A taxi must consider several possible outcomes, such as:

- Safety
- Travel time
- Traffic
- Passenger comfort
- Fuel/energy consumption

A utility-based agent can compare different actions and select the action with the best expected utility.

E1 → Utility-Based Agent

Easiest: E3 – Chess

A suitable architecture is a Goal-Based Agent.

The main goal is to win the game. The agent evaluates possible moves and chooses moves that help it achieve this goal.

E3 → Goal-Based Agent

(d) Interpretation

The complexity of an agent depends on the properties of its environment, not simply on the name of the application.

For example, an autonomous taxi operates in an environment that is partially observable, stochastic, multi-agent, dynamic and continuous. Therefore, it requires a more sophisticated agent architecture.

Chess is comparatively easier according to the given scoring system because the board is fully observable, deterministic and discrete.

Thus: More difficult environment properties → More complex agent program

3. Turing Test Data Across the History of AI

The question gives 50 interrogations for each system and asks us to calculate correct-identification and fooling rates.

(a) Correct-identification rate and fooling rate

Formula

Correct Identification Rate=Correctly identified/50×100

Fooling Rate=100−Correct Identification Rate

ELIZA

Correctly identified = 47

47/50×100=94 %

Fooling rate:

100−94=6%

Chatbot B

36/50×100=72%

Fooling rate:

100−72=28%

System C

2950×100=58%

Fooling rate:

100−58=42%

Final table

System	Correct identification	Fooling rate
ELIZA	**94%**	**6%**
Chatbot B	**72%**	**28%**
System C	**58%**	**42%**

(b) Turing's 70% criterion

Turing's reference criterion says that an average interrogator would have no more than a 70% chance of making the correct identification.

We compare each system's correct-identification rate with 70%.

ELIZA

94%−70%=24%

ELIZA is 24 percentage points above the 70% criterion.

Therefore, ELIZA does not satisfy the criterion.

Chatbot B

72%−70%=2%

Chatbot B is 2 percentage points above the criterion.

Therefore, Chatbot B does not satisfy the criterion.

System C

58%−70%=−12%

System C is 12 percentage points below the criterion.

Therefore, System C satisfies the criterion.

Distance from guessing level

A purely guessing judge is correct 50% of the time.

System	Correct rate	Difference from 50%
ELIZA	94%	+44 percentage points
Chatbot B	72%	+22 percentage points
System C	58%	+8 percentage points

Conclusion: System C comes closest to fooling the judges and has the lowest correct-identification rate.

(c) Improvement in fooling rate

ELIZA → System C

ELIZA fooling rate = 6%

System C fooling rate = 42%

Improvement:

42−6=36 percentage points

Ratio: 42/6=7

So System C's fooling rate is 7 times the fooling rate of ELIZA.

Chatbot B → System C

42−28=14 percentage points

Therefore:

- ELIZA → System C = +36 percentage points
- ELIZA → System C ratio = 7:1
- Chatbot B → System C = +14 percentage points

(d) Interpretation

The Turing Test mainly evaluates whether a machine can act humanly in a conversation. A high fooling rate means that judges sometimes mistake the machine for a human.

However, a high fooling rate does not prove that the machine is actually thinking or acting rationally.

The test evaluates the machine's observable behaviour, not its internal understanding or reasoning process.

For example, a system may generate convincing responses by recognizing language patterns without actually understanding the meaning. This is related to the Chinese Room argument, which questions whether producing appropriate language responses necessarily means genuine understanding.

Therefore: High fooling rate ≠ proof of thinking or rationality

4. Rational Route Choice Under Uncertainty

The taxi has three possible routes. Its utility is:

$U=100-T-20L$

where:

- TT = travel time
- $L=1$ if the taxi arrives after 55 minutes
- $L=0$ otherwise.

The question provides the route times and disruption probabilities.

(a) Utility of each route

Route R1 – Highway

Normal time = 30 minutes.

Since:

$30<55$

there is no late penalty.

$U=100-30-20(0)$

$U=70$

Disrupted time = 70 minutes.

Since:

$70>55$

the taxi is late, so $L=1$.

$U=100-70-20(1) =100-70-20$

$U=10$

Route R2 – City road

Normal time = 45 minutes.

$45<55$

Therefore $L=0$.

$U=100-45$

$U=55$

Disrupted time = 60 minutes.

$60>55$

Therefore $L=1$.

$U=100-60-20$

$U=20$

Route R3 – Ring road

Time = 50 minutes and there is no disruption.

$50<55$

Therefore:

U=100−50

U=50

Summary

Route	Normal utility	Disrupted utility
R1	**70**	**10 (Late)**
R2	**55**	**20 (Late)**
R3	**50**	**50**

(b) Expected utility

Formula:

EU=∑P(outcome)×U(outcome)

R1

Probability of disruption = 0.30

Probability of normal = 0.70

EU(R1) = (0.70) (70) +(0.30) (10) =49+3=49+3

EU(R1) =52

R2

Probability of disruption = 0.10

Probability of normal = 0.90

EU(R2) = (0.90) (55) +(0.10) (20) =49.5+2=49.5+2

EU(R2) =51.5

R3

No disruption:

EU(R3) =1(50)

EU(R3) =50

Comparison

Route	Expected Utility
R1	**52**
R2	**51.5**
R3	**50**

Therefore: R1 is the rational choice because it has the highest expected utility.

(c) New percept: Highway accident alert

The probability of disruption on R1 increases from 0.30 to 0.80.

Therefore:

P(normal)=1−0.80=0.20

Expected utility:

EU(R1) = (0.20) (70) +(0.80) (10) =14+8

EU(R1) =22

Now compare:

Route	Expected Utility
R1	**22**
R2	**51.5**
R3	**50**

Therefore, the rational agent should now choose: R2

The new percept changes the decision because the probability of a highway accident has increased significantly.

(d) Was the original decision irrational if disruption occurred?

In part (b), the agent selected R1 because:

EU(R1) =52

was the highest expected utility.

If the disruption actually occurred, the utility obtained was:

U=10

But this does not mean the agent was irrational.

The agent made its decision using the information available before the disruption occurred. At that time, R1 had the highest expected utility.

Rational agent vs omniscient agent

A rational agent chooses the action expected to maximize performance based on its available percepts and knowledge.

An omniscient agent would know the actual future outcome in advance.

Therefore, an omniscient agent would know that the accident would occur and would avoid R1. The rational agent did not have that future information.

An unlucky outcome does not necessarily mean an irrational decision.

(e) Interpretation

Rationality depends on:

1. **Percept sequence** – what information the agent has received.
2. **Performance measure** – what the agent is trying to maximize.
3. **Available knowledge** – the probabilities and expected outcomes known to the agent.

In part (b), the highway had the highest expected utility:

52>51.5>50

so choosing R1 was rational.

In part (c), a new percept reported a major highway accident. The disruption probability increased to 80%, reducing:

EU(R1) =22

Therefore, R2 became the best choice.

The physical roads did not change, but the agent's information and probabilities changed. Hence, the rational decision also changed.

New percepts can change the rational action even when the environment itself has not change d.

Final answers at a glance

Question	Key final result
Q1	Agent X = 6, Agent Y = 9 for p=1p=1; Y better when p > 0
Q2	Difficulty: E1 = 6, E2 = 5, E3 = 2.5; ranking E1 > E2 > E3
Q3	Fooling rates: ELIZA 6%, B 28%, C 42%; C satisfies 70% criterion
Q4	Initial choice R1 (EU=52); after alert R2 (EU=51.5)

These answers are based directly on the four questions and data provided in your uploaded assessment document.